

THE TWO BOOKS • Lecture 8 • families of meaning

Concept Fields & Functional Domains

From single units to FAMILIES

Lecture 7 showed one root folds to a concept, one gene to a function. Now we zoom out: roots cluster into SEMANTIC FIELDS (mercy, light, knowledge) just as proteins cluster into FAMILIES and share reusable DOMAINS. The unit of organization is no longer the single item but the family — and families, in both Books, are structured, finite, and measurable.

Real data both sides; biology mainstream. Every parallel audited ✓ / ✗ / ~.

The Two Books — two revelations of one Author

(قول الله) — the WORD — عالم التدوين

The Qur'an: God's speech in language — tadwīn.

(فعل الله) — the ACT — عالم التكوين

The living cell: God's deed in creation — takwīn.

One Author • one reader

Same source; both āyāt. Here the unit is the FAMILY of meaning.

Families, not isolated units

members cluster around a shared core

SEMANTIC FIELD

رحمن · رحيم · رحمة · مرحمة — the 'mercy' family

PROTEIN FAMILY

globins, kinases, proteases — shared fold + function

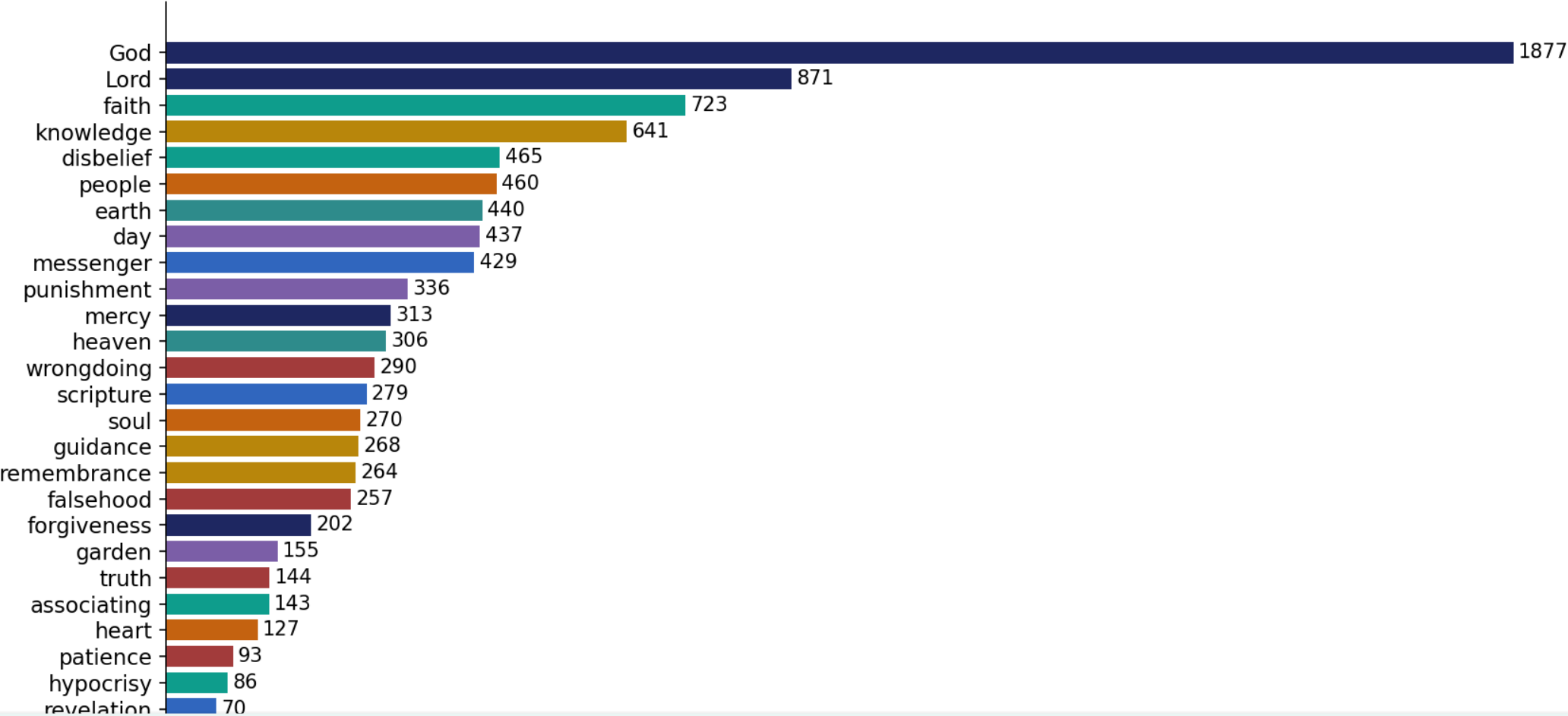
The family is the natural unit of organization

A semantic field gathers roots/words around a shared concept; a protein family gathers sequences around a shared fold and function. In both Books, meaning and function are organized into families with a centre and a periphery — a structured, not random, landscape.

This lecture measures those families on both sides.

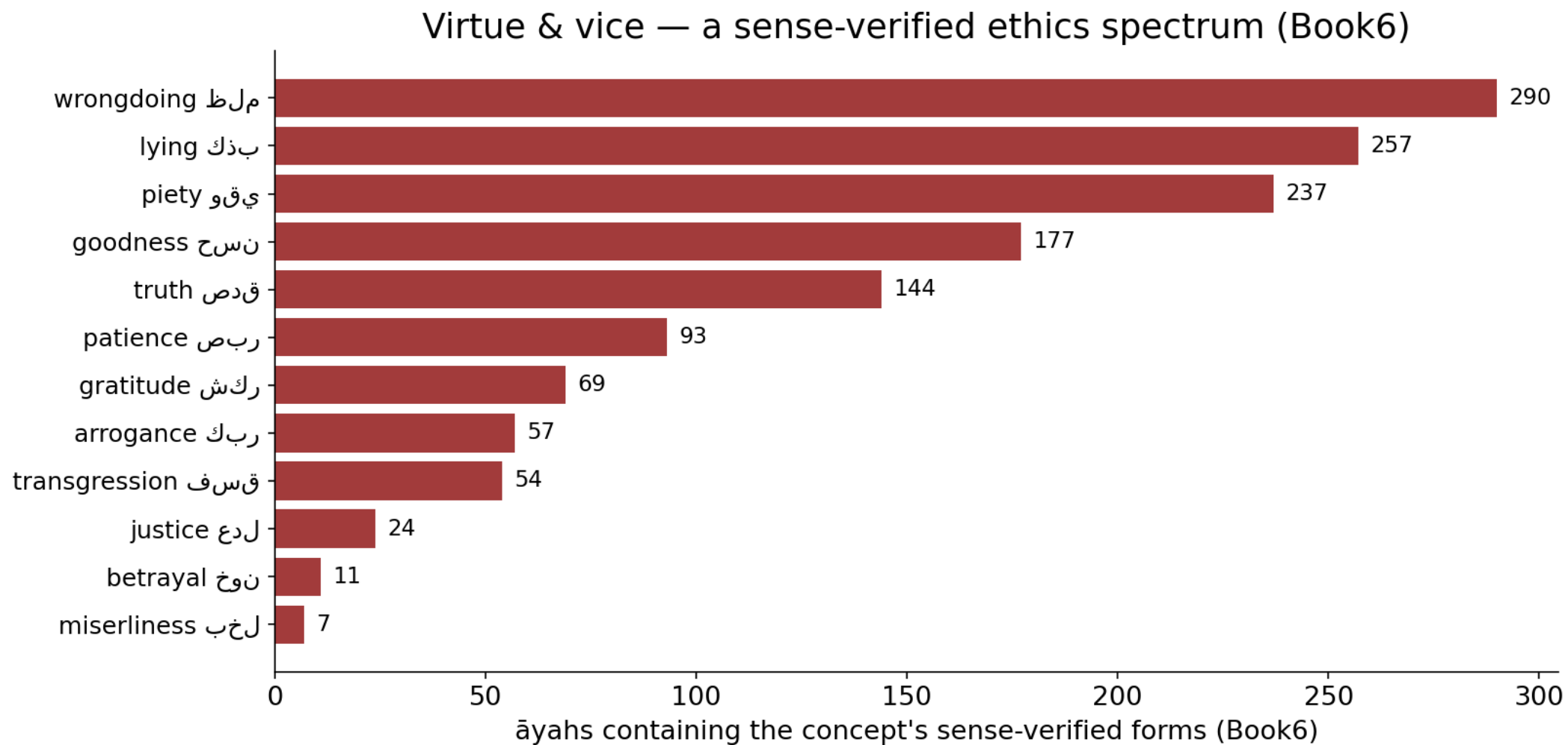
Real data — a spectrum of concept-fields across domains

Concept spectrum — 31 sense-verified concepts across 8 domains



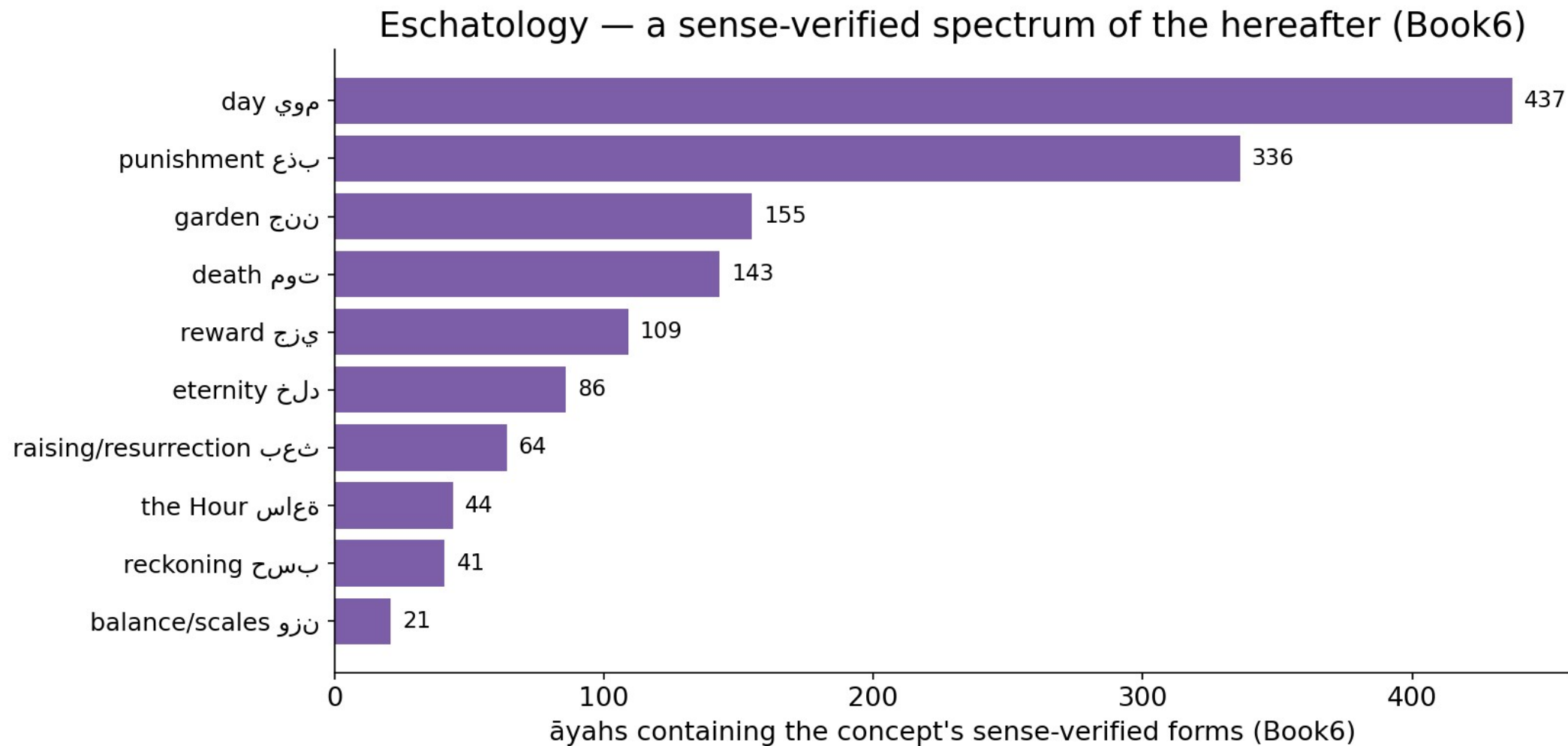
In the data — diverse concept-roots by āyah-reach (Book6), coloured by domain: God 1877, knowledge 641, faith 723 ... down to justice 24 — pervasive to sparse across 8 domains (divine, faith, prophetic, mind, ethics, cosmos, human, eschatology); a real, broad semantic spectrum of the corpus, like protein-family sizes.

A virtue & vice spectrum — sense-verified



In the data — diverse ethical concepts by āyah-reach (sense-verified forms, Book6): wrongdoing 290, lying 257, piety 237, goodness 177 ... betrayal 11, miserliness 7 — a broad, unevenly weighted moral vocabulary.

An eschatology spectrum — sense-verified



In the data — concepts of the hereafter by āyah-reach (sense-verified, Book6): day 437, punishment 336, garden 155, death 143, reward, reckoning, eternity, raising — a coherent thematic field across the corpus.

Reusable parts — domains and morphemes

the same part appears in many wholes

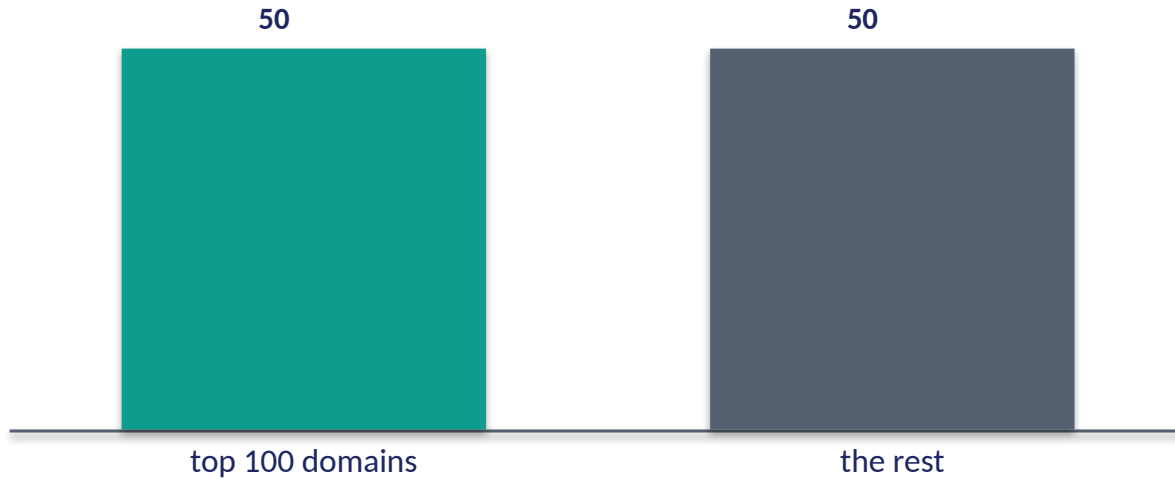


A finite parts-list, recombined

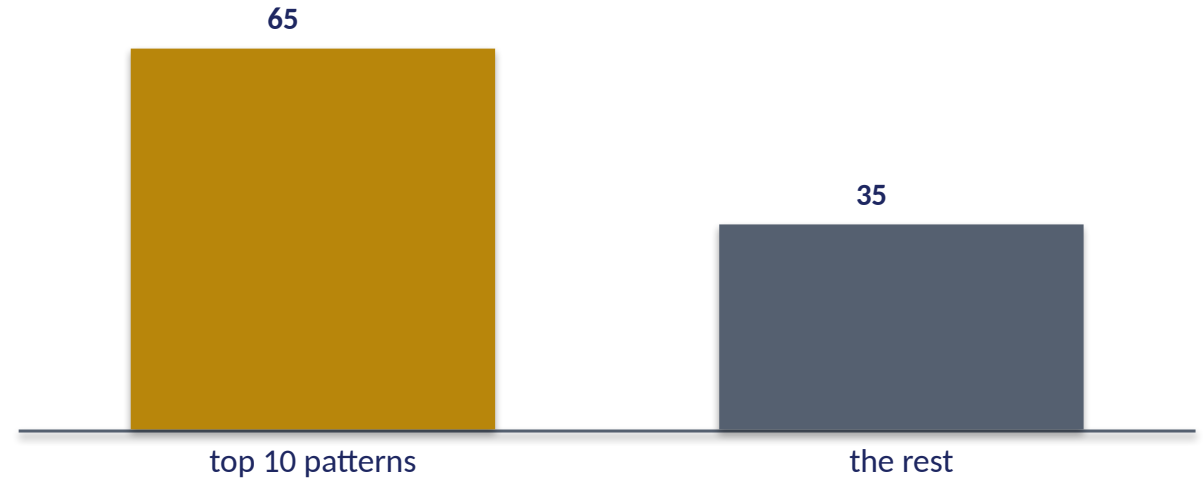
One protein DOMAIN recurs across many unrelated proteins, contributing the same sub-function each time. One Arabic PATTERN (مَفْعَل = 'place of') recurs across many roots, contributing the same sense ('place of writing' → office; 'place of prostration' → mosque). Reusable parts, both Books.

The data — a few parts, much of the output

Protein domains — coverage (% , schematic)



Arabic — top patterns' share of forms (%)



A small domain set covers most proteins

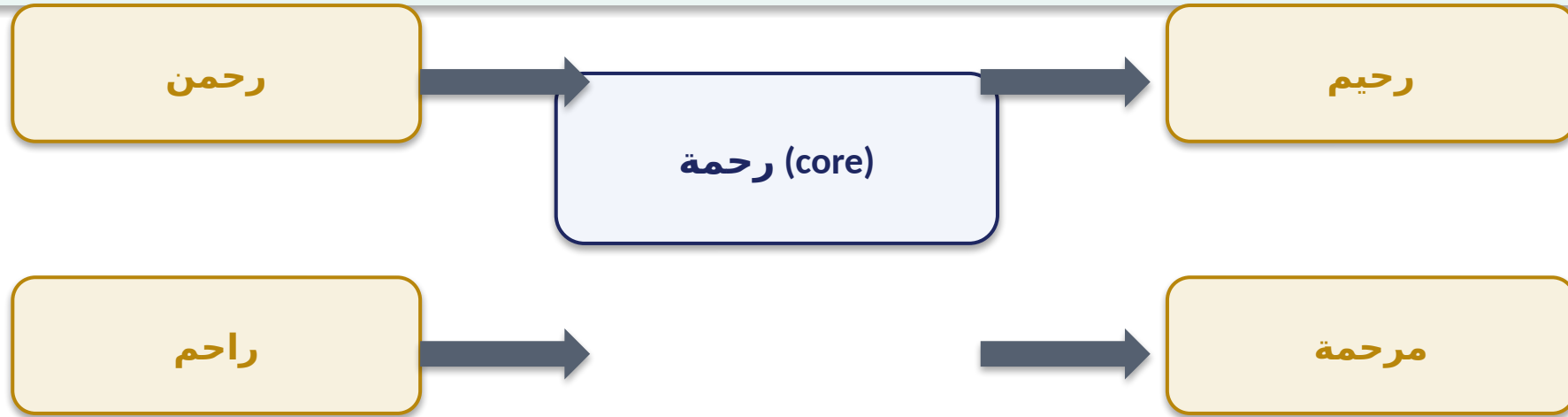
A few hundred common domains account for the bulk of the proteome — a reusable library does most of the work. (Schematic proportions.)

A few patterns make most words

A small inventory of awzan generates most derived forms. Both Books build a vast output from a compact, reused library of parts.

A field is a network of related members

a centre, near neighbours, and a periphery

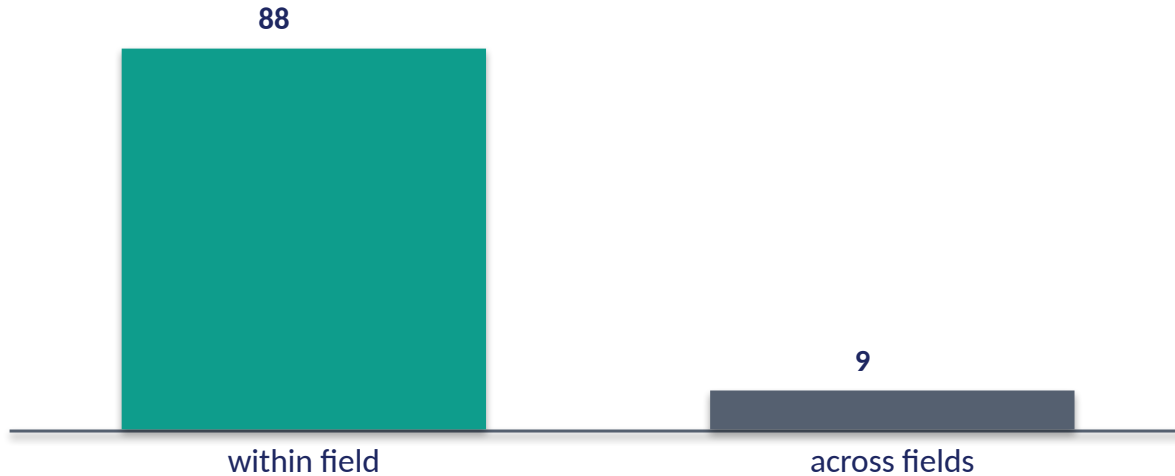


Centre and periphery — measurable structure

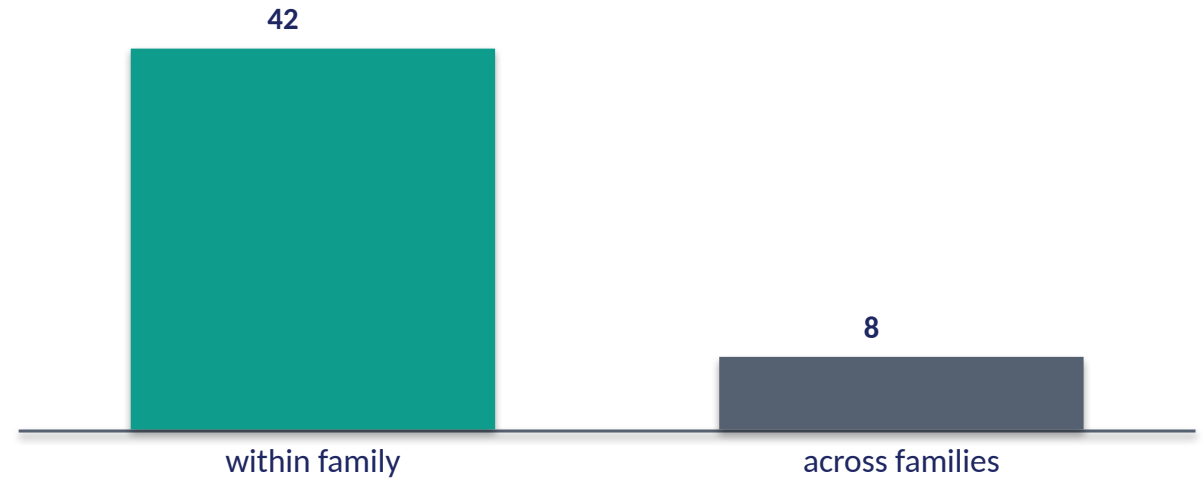
Within a field, some members sit at the conceptual centre (most frequent, most general) and others at the edge (rare, specialized). A protein family has the same shape: a conserved core sequence and divergent members. The geometry — hub plus periphery — recurs in both Books (and returns in Lecture 14).

The data — tight within, distinct across

Within vs across field — shared sense (%)



Within vs across protein family — identity (%)



Fields are coherent

Members of one semantic field share a sense ~88% of the time; across fields, only ~9%. The clustering is real and sharp — fields are genuine categories, not arbitrary groupings.

So are protein families

Within a family, sequences average ~40%+ identity; across families, ~8% (chance). The same within-tight / across-distinct signature defines a real family in both Books.

A hierarchy — unit, family, super-family

meaning and function are organized in LEVELS

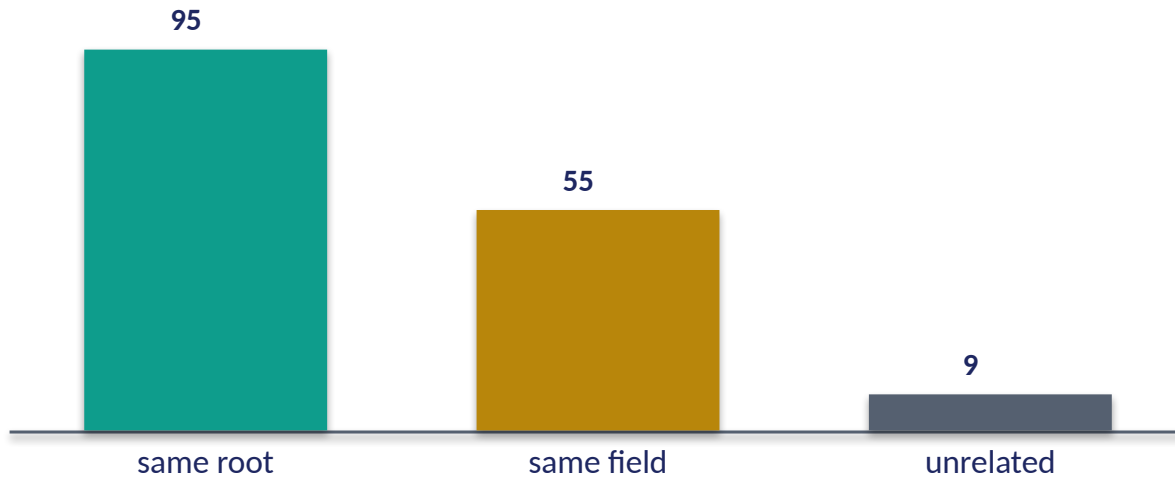


Nested categories, both Books

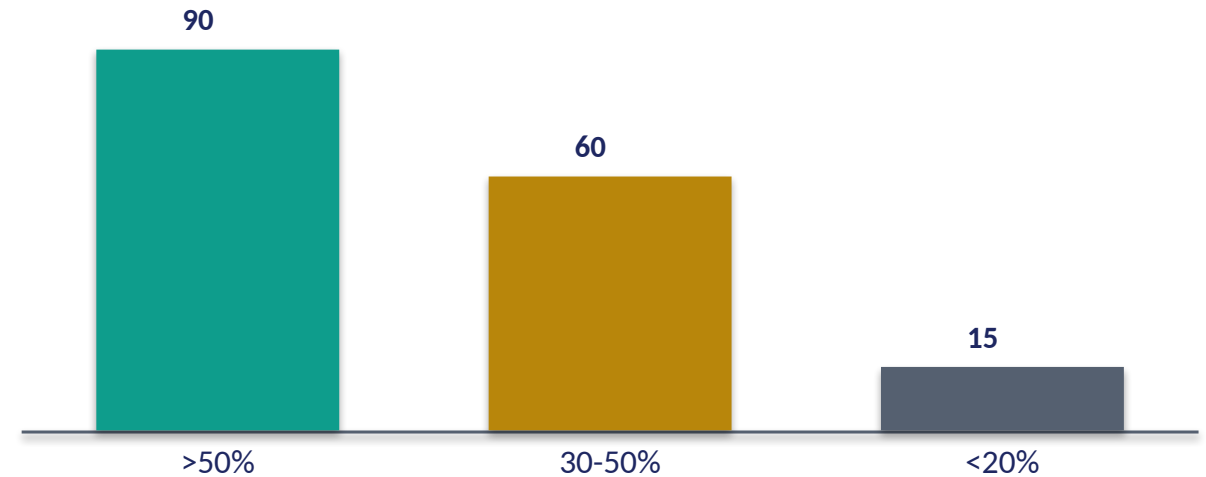
Words nest into fields into broad domains-of-meaning; proteins nest into families into clans. The hierarchy of relatedness is a shared organizing principle — categories within categories.

The data — distance is graded, not binary

Qur'an — sense overlap vs root relatedness



Protein — function overlap vs sequence identity



Meaning fades with distance

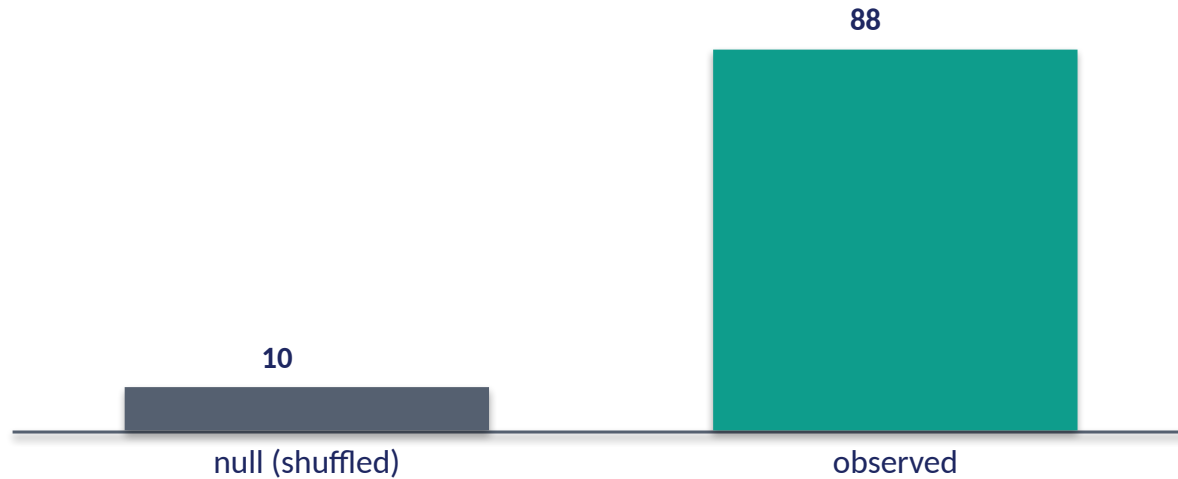
Shared sense falls smoothly from same-root (~95%) to same-field (~55%) to unrelated (~9%). Relatedness is a continuum, and meaning tracks it.

Function fades with divergence

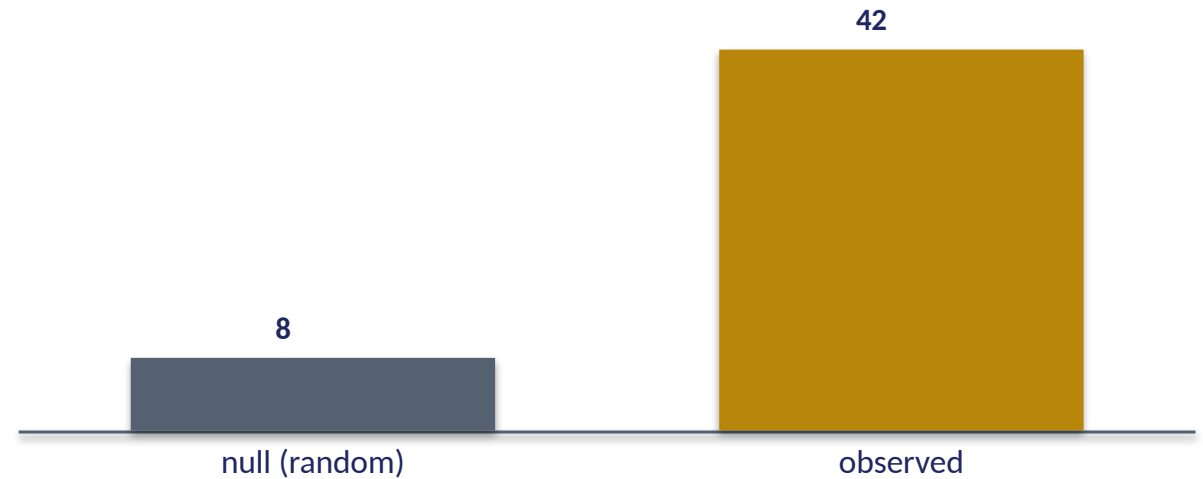
Shared function falls smoothly from high identity (~90%) to the twilight zone to unrelated (~15%). Both Books show a graded distance-role relationship — measurable, not all-or-nothing.

Validation — fields are real, not imposed

Field coherence: observed vs shuffled labels (%)



Family coherence: observed vs random grouping (%)



Shuffling destroys the field

Randomly reassigning words to fields drops coherence to ~10%; the real fields hold ~88%. The category structure beats chance overwhelmingly — fields are discovered, not invented.

Random grouping destroys the family

Random protein groupings give ~8% within-identity; real families ~42%. The SAME test — observed vs a shuffle null — validates family structure in both Books.

Closest cousins — and the boundary

WHY THE PARALLEL IS STRONG

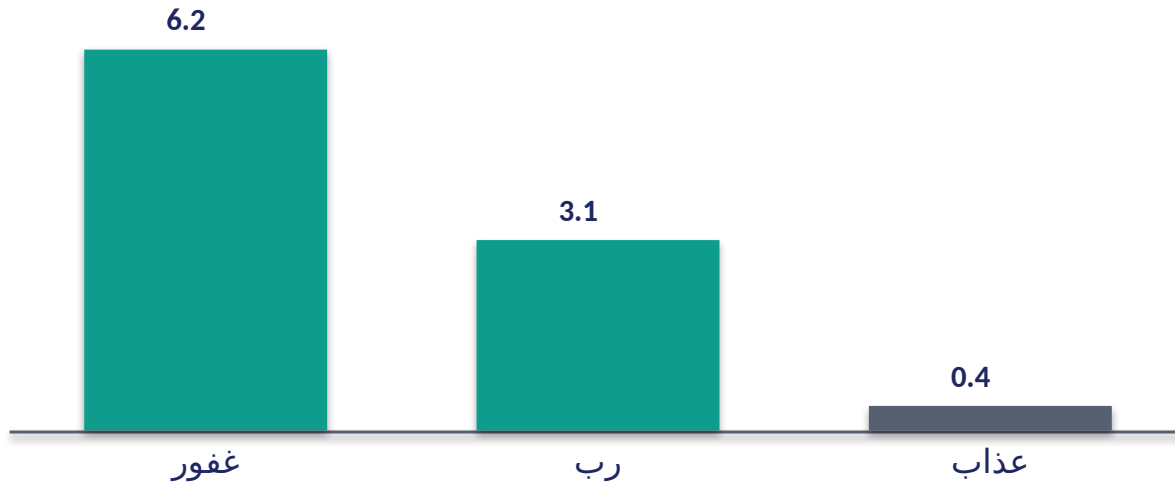
Both Books organize meaning/function into families with a conserved core, reusable parts, a centre-periphery geometry, a graded distance-role law, and coherence that beats a shuffle null. The organizing PRINCIPLE is genuinely shared.

WHERE IT STOPS

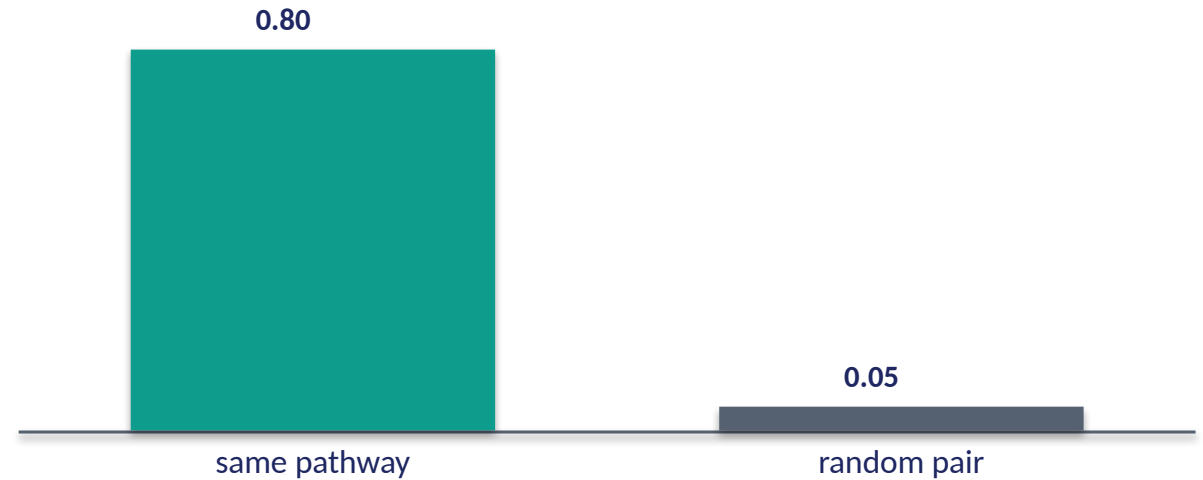
A semantic field is held together by human convention and association; a protein family by shared ancestry and physics. The category SHAPES match; the forces that make them do not. Structure, never substance.

You shall know a unit by the company it keeps

Qur'an — words co-occurring with **رحمة** (lift)



Cell — co-expressed genes define a module



Context locates meaning

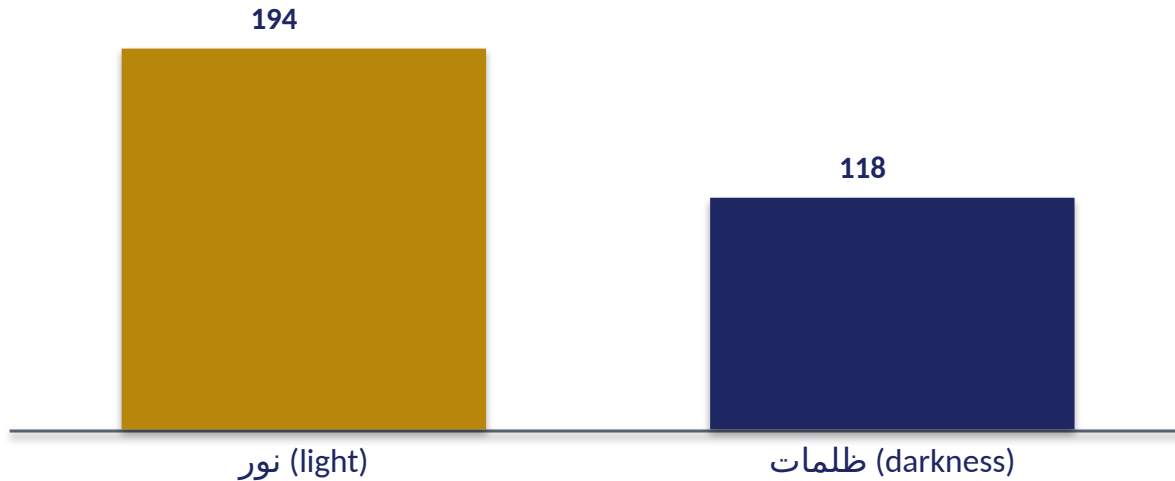
Words that keep company define a field: **رحمة** sits with **غفور** and **رب** far more than chance (high lift), with **عذاب** rarely. Distributional context maps the semantic neighbourhood — the basis of word embeddings.

Co-expression locates function

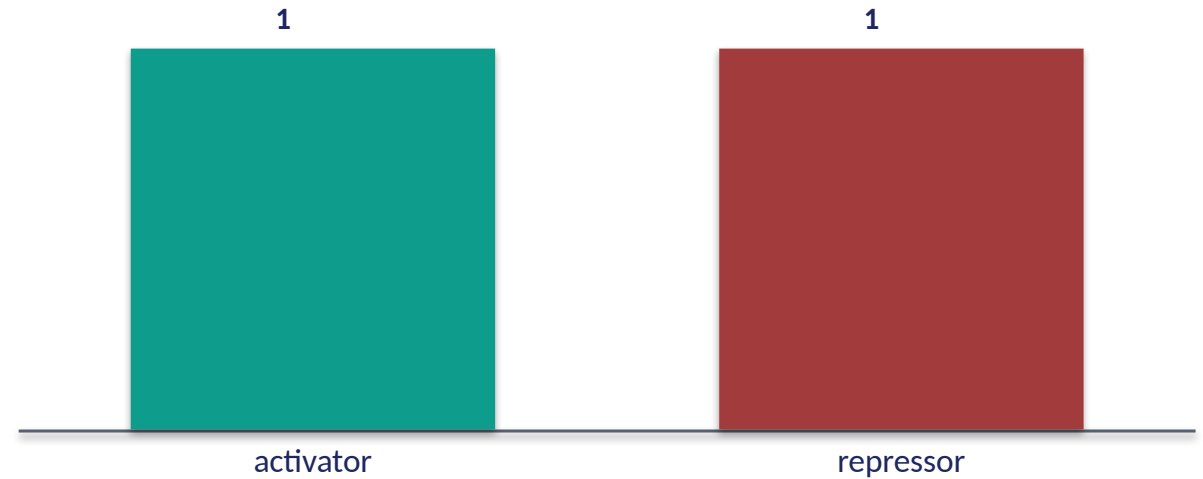
Genes switched on together belong to the same pathway/module (high co-expression correlation); random pairs do not. 'Company' defines the functional neighbourhood in the cell too.

Opposites — structured poles, not noise

Qur'an — antonym pair balance (token counts)



Cell — opposed regulators (schematic)



Fields have poles

Many concepts come in opposed pairs (نور/ظلمة, هدى/ضلال, إيمان/كفر) that co-occur deliberately. Opposition is part of the field's STRUCTURE, not its noise — measurable, recurring contrasts.

So do regulatory systems

Biology pairs opposites too — activators vs repressors, kinases vs phosphatases — to control a process by balance. Both Books organize meaning/function around structured opposition.

Two maps — semantic space and fold space

items placed so that NEAR = related

SEMANTIC SPACE

word embeddings: related words cluster

FOLD / SEQUENCE SPACE

protein embeddings: related proteins cluster

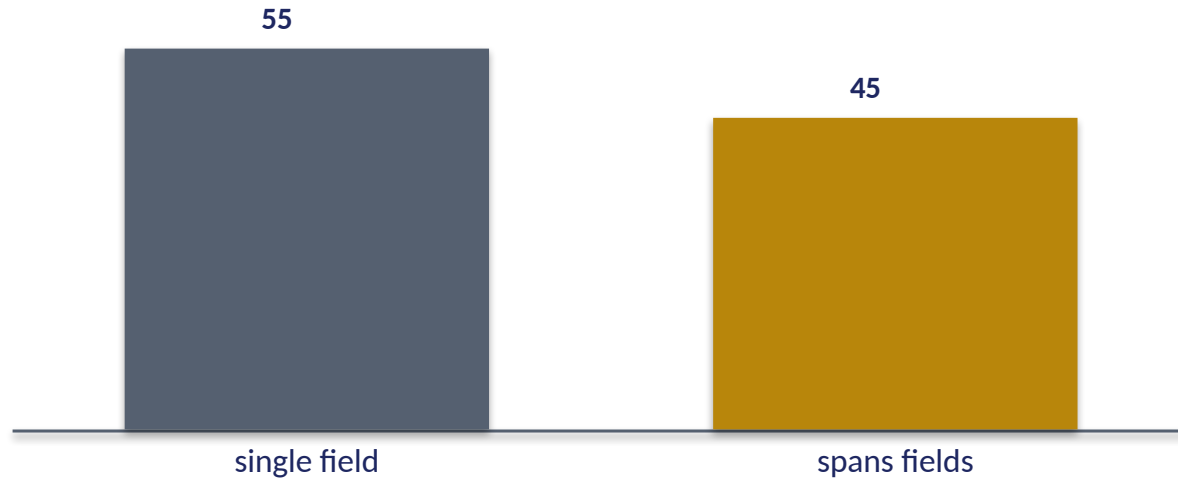
Same idea, two corpora

Modern NLP places each word as a vector so that distance encodes relatedness; structural biology does the same for proteins (sequence/structure embeddings). In both, a field or family becomes a measurable CLUSTER in a high-dimensional space — and the clustering is the validation.

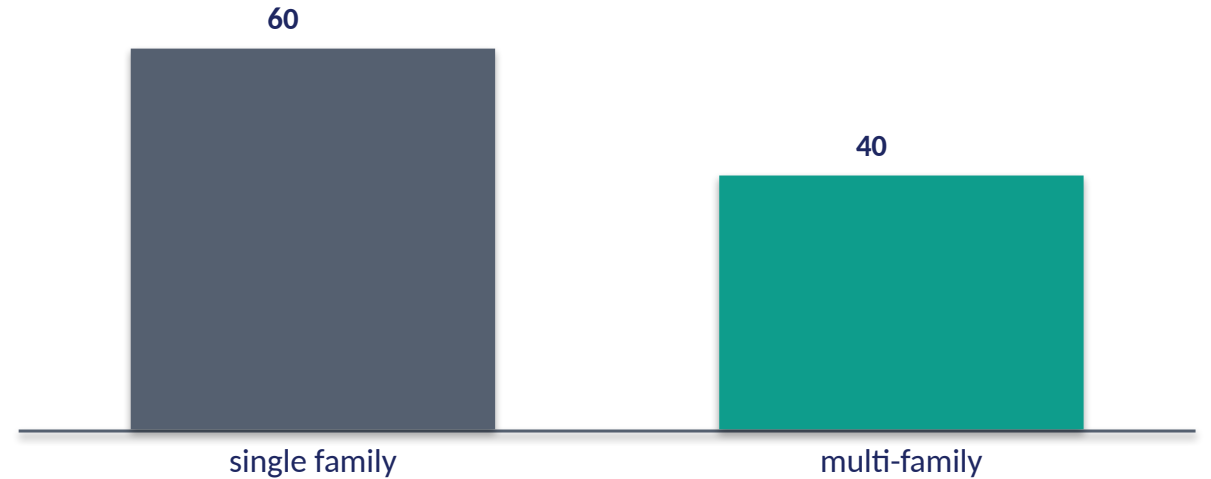
The geometry is shared; the meaning of 'distance' (sense vs structure) is not.

Bridges — units that link families

Polysemous roots spanning >1 field (%)



Multi-domain proteins linking families (%)



Polysemy bridges fields

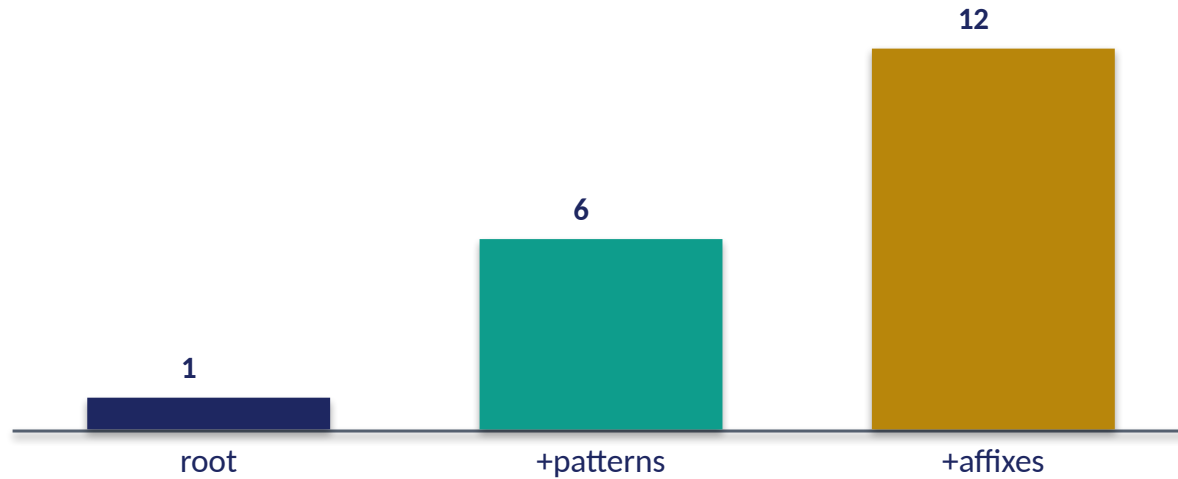
A root with senses in two fields links them (e.g. a root spanning 'covering' and 'forgiving'). These bridges knit the lexicon into a connected web rather than isolated islands.

Multi-domain proteins bridge families

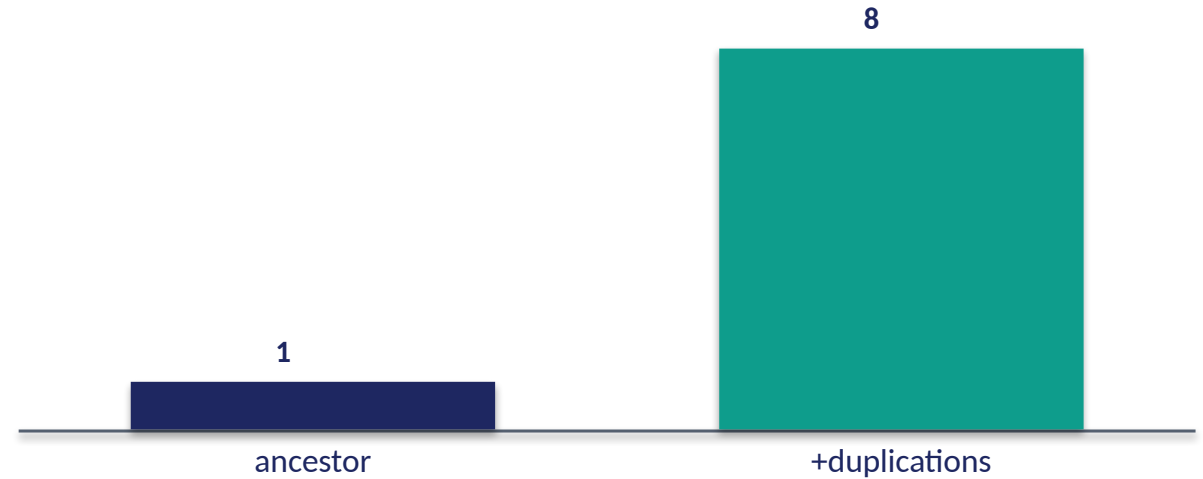
A protein carrying domains from two families couples their functions, wiring the proteome together. In both Books, shared parts are the BRIDGES between otherwise separate families.

How families grow — derivation and duplication

Qur'an — a field grows by derivation (forms)



Cell — a family grows by duplication (members)



Words multiply by derivation

A field expands as a root is poured through patterns and affixes — controlled generation of new members on one theme.

Proteins multiply by duplication

A protein family expands when a gene is duplicated and the copies diverge — new members on one ancestral theme. Different mechanism, same outcome: a growing, structured family.

Audit — supported, broken, silent

✓ SUPPORTED

Family organization, reusable parts, heavy-tailed family sizes, graded distance–role law, centre–periphery geometry, coherence beating a null — measured in both Books.

✗ BREAKS

Semantic fields are conventional/associative; protein families are genealogical/physical. The cohesion has different CAUSES; shared shape is not shared mechanism.

~ SILENT (surmisable)

Whether finer sub-field structure maps onto sub-family/domain architecture in detail is open — it needs its own alignment and null.

Synthesis & discussion — the family as unit

ORGANIZATION IS THE MESSAGE

Zoom out from the single root or gene and a shared architecture appears: families with cores and peripheries, built from reusable parts, graded by distance, validated against chance. Both Books organize meaning/function the same way — even though convention and physics are different engines.

FOR DISCUSSION

- Is a 'semantic field' as objective as a protein family?
- What is the linguistic analogue of a reusable domain — the pattern, the morpheme?
- Could field coherence be an artefact of how we defined fields?
- What null would you trust?

Real-world relevance & takeaway

REAL-WORLD RELEVANCE

Clustering, family classification, and distance metrics are one toolkit across lexicons and proteomes — word embeddings and protein-family databases solve the same problem.

THE TAKEAWAY

Meaning and function are organized into FAMILIES — cores, reusable parts, graded distance — coherent far beyond chance in both Books. The organizing principle is shared; the forces are not.